Worked example SL12: Zhang Cong's titles and a disputed date

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Purpose of this example

This is the second worked example. SL14 showed the whole procedure on a short passage. SL12 is longer (58 information units from three passages) and was the pilot case, so it is presented differently: the overall results are summarised, and three problems are followed in detail, chosen because they concern disagreement and interpretation, the subject of RQ2:

1. a date on which four sources disagree;
2. an error in one source, confirmed on the page image;
3. a phrase, "his other offices as before", that implies some titles were replaced without saying which.

Two things are different from SL14 and should be kept in mind. First, SL12 was the pilot: its information units were listed *after* the first version of the event model had been written, not before. Second, the event model was revised after its first evaluation, and both evaluations are reported. Everything below follows the frozen evaluation (event model v2; state model eval-state-v1).

As in SL14, office and title names follow the English labels in the project vocabulary, which use Hucker (1985) where he has an entry; for the two hall titles Hucker has no entry, so the pinyin is kept: "Grand Secretary of the Huagai Hall" is *Huagaidian Daxueshi* 華蓋殿大學士 in the model, and similarly for the Jinshen Hall.

The source passages

All three passages are from the *Ming Shizong shilu*. They are quoted as transcribed in the project's working files (simplified characters; punctuation of the platform's edition). All three have been checked against the page images; where the transcription has 升, the images read 陞.

Juan 89, Jiajing 7/6 *xinchou*, first day of the month (27 June 1528).

> 嘉靖七年六月辛丑朔，明伦大典书成进呈，上亲制序文，命宣付史馆，刊布天下。加恩纂述效劳诸臣。少师兼太子太师、吏部尚书、华盖殿大学士杨一清加正一品俸，荫一子为尚宝司司丞。少保、礼部尚书兼文渊阁大学士张璁加少傅、太子太傅，升吏部尚书、谨身殿大学士，荫一子为中书舍人。

The *Minglun dadian* is presented; the compilers are rewarded. Zhang Cong, Junior Guardian and Minister of Rites concurrently Grand Secretary of the Hall of Literary Profundity (文淵閣), is *added* (加) the titles Junior Mentor and Grand Mentor of the Heir Apparent, and *promoted* (陞) to Minister of Personnel and Grand Secretary of the Jinshen Hall.

Juan 136, Jiajing 11/3 *jiayin* (20 April 1532).

> 甲寅，原任太傅兼太子太傅、吏部尚书、谨身殿大学士张孚敬应召至京，诏进兼太子太师、华盖殿大学士，余官如故。孚敬疏辞。上曰：卿四郊督视，克尽勤诚，辅赞朕躬，多赖匡翊。兹加恩非过典，其即承朕命，尽心供职，勿辞。

Zhang Fujing (the name bestowed on Zhang Cong in 1531), arriving at the capital in answer to a summons, is *advanced* (進) to Grand Preceptor of the Heir Apparent and Grand Secretary of the Huagai Hall, "his other offices as before". He declines; the emperor refuses, giving his reasons.

Juan 149, Jiajing 12/4 *yihai* (6 May 1533).

> 致仕少傅兼太子太师、吏部尚书、华盖殿大学士张孚敬赴台，至京……

The retired Zhang Fujing arrives at the capital again. Only his full title string is given; The page image reads 赴台, which in context is probably an error for 赴召 ("answering a summons"); the text is kept as written.

Two further sources enter the case only for the disputed date: the table of Grand Secretaries (*Mingshi*, juan 110) and Zhang Cong's biography (*Mingshi*, juan 196).

Information units

The three passages give 64 units, of which 58 are in scope (25 from juan 89, 25 from juan 136, 14 from juan 149, before scope). Out of scope are, for example, the emperor's own preface to the book, the privileges granted with the titles (a higher salary grade, protection privilege for a son), the form of address 卿, and the emperor's remark that the honour was "not excessive" (非過典). One unit of juan 149 was set out of scope after the page image showed that it belongs to the next entry.

The units most relevant to the three problems discussed below are these:

Unit	Type	Text	Note
J136-U01	date	甲寅	day of the conferral; year and month from the context of the juan
J136-U04	title	太傅 (Grand Mentor)	as written; juan 89 has 少傅 (Junior Mentor)
J136-U12	action	進 (advanced)	
J136-U14	title	太子太師 (Grand Preceptor of the Heir Apparent)	newly conferred
J136-U15	title	華蓋殿大學士 (Grand Secretary of the Huagai Hall)	newly conferred; the date of this unit is disputed
J136-U16	relation	餘官如故 ("his other offices as before")	which earlier titles were replaced is not stated
J136-U21	motive	四郊督視……多賴匡翊	the emperor's reason for the grant
J089-U23	relation	earlier and new titles	whether Junior Guardian was replaced by Junior Mentor is not stated

The full list is in `data/evaluation/info_units.csv`.

Problem 1: four sources, four dates

When did Zhang Cong become Grand Secretary of the Huagai Hall? Four sources give four answers:

Source	What it says	As a Gregorian range
Shilu, juan 136	Jiajing 11, third month, day *jiayin*	20 April 1532
Mingshi, juan 110 (table of Grand Secretaries)	Jiajing 11, third month	16 April – 14 May 1532
Mingshi, juan 196 (biography)	Jiajing 13	24 January 1534 – 11 February 1535

Source	What it says	As a Gregorian range
CBDB, posting 421880	1532 (coded as Jiajing 11)	16 February 1532 – 3 February 1533

The first two agree. The biography's date is contradicted by the *Veritable Records*: juan 141 already styles him Grand Secretary of the Huagai Hall when he retires in the eighth month of Jiajing 11, and juan 149 does so again in Jiajing 12, both before the biography's date. The biography's sentence (明年進少師兼太子太師、華蓋殿大學士) seems to run together his promotion to Junior Preceptor (少師), which may well belong to Jiajing 13, with titles he had received earlier. CBDB's record gives 1532 but cites the biography as its source, which says Jiajing 13; record and cited source therefore do not match.

How each representation handles it.

CBDB holds one value only. The other sources, and the fact that they disagree, cannot be seen. CBDB could in principle hold a second posting citing another source, but that was not done.

The event model (decision D7) does not give the conferral itself a date. Instead, each source's statement is recorded as a separate claim, an E13 Attribute Assignment, which says that a given time-span was assigned to the conferral, on the basis of a given source, by a given interpreter (here myself, as the modeller who read the sources):

```
# The conferral itself carries no P4 (date)
event:huagaidian-conferral a crm:E7_Activity ;
    crm:P14_carried_out_by person:ming-shizong ;
    crm:P11_had_participant person:68266 ;
    crm:P9_consists_of event:huagaidian-conferral-1 , event:huagaidian-
conferral-2 .

# Claim 3 of 4: the date according to the Mingshi biography
event:huagaidian-claim-mingshi-196 a crm:E13_Attribute_Assignment ;
    crm:P140_assigned_attribute_to event:huagaidian-conferral ;
    crm:P177_assigned_property_of_type crm:P4_has_time-span ;
    crm:P141_assigned ts:jj13 ;                    # Jiajing 13, whole
lunar year
    crm:P16_used_specific_object doc:mingshi-196 ;
    crm:P14_carried_out_by person:yang .           # the interpreter
```

The other three claims have the same form. A query can therefore return all four dates with their sources. What the model does not hold is the *reasoning* that rejects the biography: E13 records claims, not arguments. That reasoning stays in the documentation (see CQ4 below).

The state model has no time at all, so the question cannot be asked of it.

Unit J136-U15 (Huagai Hall title)	CBDB	CRM event	CRM state
Judgement	kept (but the cited source does not match the date)	kept_ext (the title through the extension property; the date through the four claims)	kept_ext (title only)

Problem 2: an error in the source

Juan 136 calls Zhang Fujing 原任**太傅**, "formerly serving Grand Mentor". Every other source calls him Junior Mentor (少傅): juan 89 records the conferral of Junior Mentor in 1528, and juans 132, 141 and 149 use Junior Mentor. Moreover, the treatise on offices states that civil officials received one of the Three Dukes (三公), to which Grand Mentor belongs, only posthumously (惟贈乃得之): Yang Tinghe was granted Grand Mentor in 1523 but declined it, and the only later exception was Zhang Juzheng in 1581–1582 (*Mingshi*, juan 72). Zhang Cong himself received one of the Three Dukes, Grand Preceptor (太師), only after his death (*Mingshi*, juan 196). I checked the reading against the page image: the manuscript does read 太傅, so the error is in the source, not in the transcription. It is an error in the *Veritable Records* as they survive, not a real title; whether it was made by the compilers or by a later copyist cannot be told.

The principle followed is that the source text is kept as it is written, errors included, and the judgement about it is recorded as a claim. In the information units, J136-U04 records 太傅 as written, with a note.

Unit J136-U04 (太傅 as written)	CBDB	CRM event	CRM state
Judgement	lost / omission	lost / omission	kept_ext

CBDB has only Junior Mentor; the variant reading is invisible. It could have been recorded as a separate posting citing juan 136.

The event model also loses it. The verbatim title string of juan 136 was not created, and there is no type for Grand Mentor. This is an omission: the same pattern is used elsewhere in the model.

The state model keeps it, as two competing claims (E13): juan 136 says Grand Mentor, juan 132 says Junior Mentor, each with its source. It also keeps the verbatim title string of juan 136.

This is the one place in SL12 where the state model clearly does better than the event model, and the reason must be stated plainly. The state model was written *after* the event model had been evaluated, and its author (myself) had seen that this unit was lost. The difference is therefore not evidence that a state-based model handles variant readings better; it shows that the same CRM pattern (E13 claims) works in both, and that the event model omitted it.

Problem 3: "his other offices as before"

In 1532 Zhang Fujing was advanced to Grand Preceptor of the Heir Apparent and Grand Secretary of the Huagai Hall, "his other offices as before" (餘官如故, unit J136-U16). The phrase implies that the new titles replaced some old ones, presumably Grand Mentor of the Heir Apparent (replaced by the higher Grand Preceptor in the same series) and Grand Secretary of the Jinshen Hall (replaced by the Huagai Hall), while the rest continued. But the text does not say which. Juan 149, a year later, supports this reading: his title string there has Grand Preceptor and Huagai Hall, and no longer the replaced titles.

The project's rule for this (decision D12) is that within one series of titles, a later title replaces an earlier one. Whether this always held in the Ming is a question for historians (Q7), so the rule is applied as the modeller's inference and not as a fact stated by the source.

Unit J136-U16 ("other offices as before")	CBDB	CRM event	CRM state
Judgement	**distorted** / omission	kept_judgement	lost / limit

CBDB partly reflects the replacement, since the postings for Grand Mentor of the Heir Apparent and the Jinshen Hall end in 1532. But the Minister of Personnel posting also ends in 1532, and a new one starts in 1533. A reader of the database would conclude that Zhang Cong was not Minister of Personnel for part of 1532–1533, which is the opposite of "as before". This is why the unit is judged *distorted*, not partial. A single continuous posting could have been kept, so the cause is omission.

The event model adds no triple for the phrase. With the D12 rule, the replacement can be inferred from the two conferrals and the series each title belongs to, so the unit is kept, but only by judgement.

The state model lists the titles held, without time. It cannot express that one title replaced another. Read carelessly, as a list of titles held at the same moment, it would even suggest that Zhang held three hall titles at once. That reading is prevented only by the comment

on the extension property, which says "held at some time".

A related unit, J089-U23, asks the same question for 1528: did Junior Mentor replace Junior Guardian? The source does not say. Here the event model scores only *partial*, because Junior Guardian, his title before 1528, was never linked to him in the model (an omission noted in the evaluation).

Results for the whole case

58 units	CBDB	CRM event (v2)	CRM state
kept / kept_ext / kept_judgement	21 / – / –	18 / 16 / 8	9 / 17 / 0
notes_only / partial / distorted	2 / 8 / 3	0 / 7 / 0	0 / 6 / 0
lost / not_modelled	24 / –	4 / 5	21 / 5
cause limit (of which my conventions)	13	2 (2)	26 (18)
cause omission	19	8	1
cause scope	5	6	5

The pattern is the same as in the other two cases. CBDB loses many units, but more of its losses are omissions than limits. The event model keeps most units, but 24 of its 42 successes depend on an extension property or on my judgement. The state model loses nearly as many units as CBDB, and almost all of its losses are limits, most of them caused by my own conventions, above all the decision to give it no events.

The revision of the event model (v1 → v2). SL12 was the pilot case, and its event model was revised after the first evaluation. Three decisions were added (D12, replacement within a series; D13, one sub-event for each title conferred; D14, Gregorian dates) and a missing block on the retirement and recall of 1531 was written. The model was then evaluated again. Ten judgements changed: five because something that had simply been left out was added, three because of D13, two because of D12. Lost units fell from 10 to 4. Both versions are kept under their own tags (`eval-v1.1` and `eval-v2`), and every change is listed in `event_v1_v2_diff.csv` with its reason. The evaluation rules were frozen only after this revision, on 28 October; the later cases SL11 and SL14 were modelled and evaluated once under the frozen rules, apart from corrections of evaluation errors that are documented.

Competency question CQ4

CQ4. When was Zhang Cong given the title of Grand Secretary of the Huagai Hall, and what does each source say? The gold answer, written from the sources before any query: juan 136, Jiajing 11/3 *jiayin*; table of Grand Secretaries, Jiajing 11/3; biography, Jiajing 13; CBDB 421880, 1532 (citing the biography, which does not match). Judgement: the *Veritable Records* and the table agree; the biography's date is later than juan 149, which already uses the title, and can be ruled out.

	Verdict	What the query returned
CBDB	wrong	Two postings: 421880, 1532, citing the biography; 421883, 1533–1535, citing juan 149. A reader would conclude that the biography says 1532 and the *Veritable Records* say 1533. Both conclusions contradict the sources
CRM event	partly correct	All four claims with their dates and sources, and the note that CBDB's date does not match its source. The judgement between them is not in the model
CRM state	cannot answer	Zhang held the title; nothing about when, and none of the four claims

The event model's result is "partly correct" because of a rule I set while reviewing the first-pass verdicts (it changed this verdict from "correct", and the change is recorded): where the gold answer contains a decision between sources, a full answer must give that decision. E13 can record what each source claims, but not why one claim should be preferred. A pattern that records arguments, such as CRMinf, or an interpretation layer such as HiCO, would be needed for that. This is one of the points I would most like to discuss.

What this example shows

- **Disagreement between sources can be represented in CIDOC-CRM without extensions.** Recording each source's date as a separate E13 claim keeps all four dates, their sources and the interpreter, and a query returns them together. CBDB, in its current data, shows a single date, and in this case a date that does not match its own cited source.

- **What E13 does not hold is the argument.** Why the biography is wrong (juans 141 and 149 already use the title) is part of the historian's work, and it remains outside the model. This is where an interpretation layer would be needed.
- **Keeping the text as written matters.** The error in juan 136 is evidence about the transmission of the source. It is lost wherever only normalised titles are recorded, and kept wherever the verbatim title string and a claim about it are recorded.
- **Inferences should be visible as inferences.** "His other offices as before" can only be turned into data with a rule about replacement that the source does not state. Counting such units as *kept_judgement*, rather than *kept*, keeps the modeller's contribution separate from what the standard and the source provide.
- **The pilot was not blind, and the results say so.** The units were listed after the first model; the model was revised after the first evaluation; and the state model was written after the event model had been evaluated, which shows in Problem 2. These are the reasons the later cases were done in a fixed order with the rules frozen.

Open points. Two known problems of the event model surfaced through the competency questions, and both are in this case.

- *CQ1.* The Minister of Personnel title of a Grand Secretary is typed as a "concurrent rank title". The type therefore asserts that the holder did not run the ministry, which the frozen version had not established. A later check of the table of ministers (*Mingshi*, juan 112) showed that the ministry was formally headed by others throughout his years in the Cabinet, and that the table, which does record Grand Secretaries who ran a ministry, never lists him (whether anyone acted for the ministry during its vacancy of 1530–1531 is not known). The design problem remains: the same title could be a rank or an office actually held. A new decision (D27) keeps titles apart from functions: the type only names the title, and whether a holder served or ran the office is recorded for each case, with its source.
- *CQ3.* The dismissal and recall of 1529 were left out of scope, so a query infers that Zhang Cong was continuously a member of the Cabinet.

Both are recorded and left unchanged in the frozen evaluation, to be corrected in a separately reported revision.

Where to find everything in the repository

Repository: github.com/KikaYang/ming-prosopography-crm. It is a snapshot of the work at this stage; the version tags mentioned here (such as `eval-v1.1` and `eval-v2`) belong to the full development history, which is kept privately and is available on request.

What	File
Information units (rows J089, J136, J149)	`data/evaluation/info_units.csv`
Event model, blocks ①–⑩, with diagram	`docs/modelling/SL12_event.ttl`, `docs/modelling/diagrams/`
State model	`docs/modelling/SL12_state.ttl`
The four date claims (Problem 1)	`SL12_event.ttl`, block ⑨
The Grand Mentor claims (Problem 2)	`SL12_state.ttl`
Decisions D1–D15 used here, Q5 and Q7	`docs/modelling/modelling_decisions_en.md` (English)
Evaluation tables	`data/evaluation/cbdb_vs_units.csv`, `event_vs_units.csv`, `state_vs_units.csv`
Changes from v1 to v2 of the event model	`data/evaluation/event_v1_v2_diff.csv`
Evaluation rules and version history	`data/evaluation/README_en.md` (English)
CQ4 query, raw results and verdicts	`data/evaluation/cq_queries/CQ4*`, `cq_raw/`, `cq_results.csv`
CBDB records for Zhang Cong (person 68266)	`data/deep_cases/cbdb_postings_3persons.csv`